

Introduction to Computer and Computer Science

Homework Assignment #8

Due: 11/25 0:00 am

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Numerals

Numerals are one of the most important symbols in the history of mankind. They have been widely used in many parts of your life such as currency systems, mathematics, religions, About 2000 years ago, the Roman Empire also had their numeral system to represent numbers. Unlike the current numeral system based on Arabic numerals, the system of Roman numerals uses special characters, I (1), V (5), X (10), L (50), C (100), D (500), M (1000), ... etc. to represent different values of numbers, with a rigorous rule.

In this homework, we are going to develop a program, which aims to transform an arbitrary Arabic numeral into Roman numeral. For simplicity, we only consider positive integers less than 4000 in this homework since it is inconvenient to type the special characters like $\bar{V}$ (5000), $\bar{X}$ (10000), $\bar{L}$ (50000), ... in Python. In addition, you have to create some rules to tell Python how to deal with the operations like addition, subtraction and comparison of Roman numerals. You will need to use “class” to accomplish this task.

Problem #1: Roman numerals

Please create a class object called “**numerals**” which can represent either the Arabic or Roman numerals and meet the specification as shown below.

Let assume we have,

```
n1 = numerals(1800),  
n2 = numerals(1),  
n3 = numerals(666),  
n4 = numerals(9487).
```

After you run the code as shown below,

```
51 print(n1)  
52 print(n1 + n2)  
53 print(n1 - n3)  
54 print(n1 + n4)
```

your program should be able to return you these messages,

```
MDCCC
MDCCCI
MCXXXIV
I don't have enough symbols to represent the number: 11287
```

To summarize these test cases,

Object 1: $n1 = \text{numerals}(1800)$

Object 2: $n2 = \text{numerals}(1)$

Object 3: $n3 = \text{numerals}(666)$

Object 4: $n4 = \text{numerals}(9487)$

Case 1

Operation: $\text{print}(n1)$

And the program should return you: MDCCC

Case 2

Operation: $\text{print}(n1 + n2)$

And the program should return you: MDCCCI

Case 3

Operation: $\text{print}(n1 - n2)$

And the program should return you: MCXXXIV

Case 4

Operation: $\text{print}(n1 + n2)$

Should return you: I don't have enough symbols to represent the number: 11287

Case 5

Operation: $n3.\text{decimal}()$

And the program should return you: 666

Case 6

Operation: $n3 > n1$

And the program should return you: False

Please be aware, the teaching assistants will use “import” command to import your program and execute some testing code to examine your program. **The name of your class object, including the name of corresponding methods, should be exactly as same as the name we shown here.** Otherwise it will pop out an error and terminate the testing procedure. In that case, your homework will be considered to be failed. **Please double check your program carefully.**

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```

6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |

```

Please save your code as a “.py” file, where the file name should follow this format:

108A_**hw#?**_**YourID**.py

For example,

108A_**hw#1**_**0180506**.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.